LEVEL DESIGN BLUEPRINT

WAREHOUSE: 3-LANE TDM ARENA

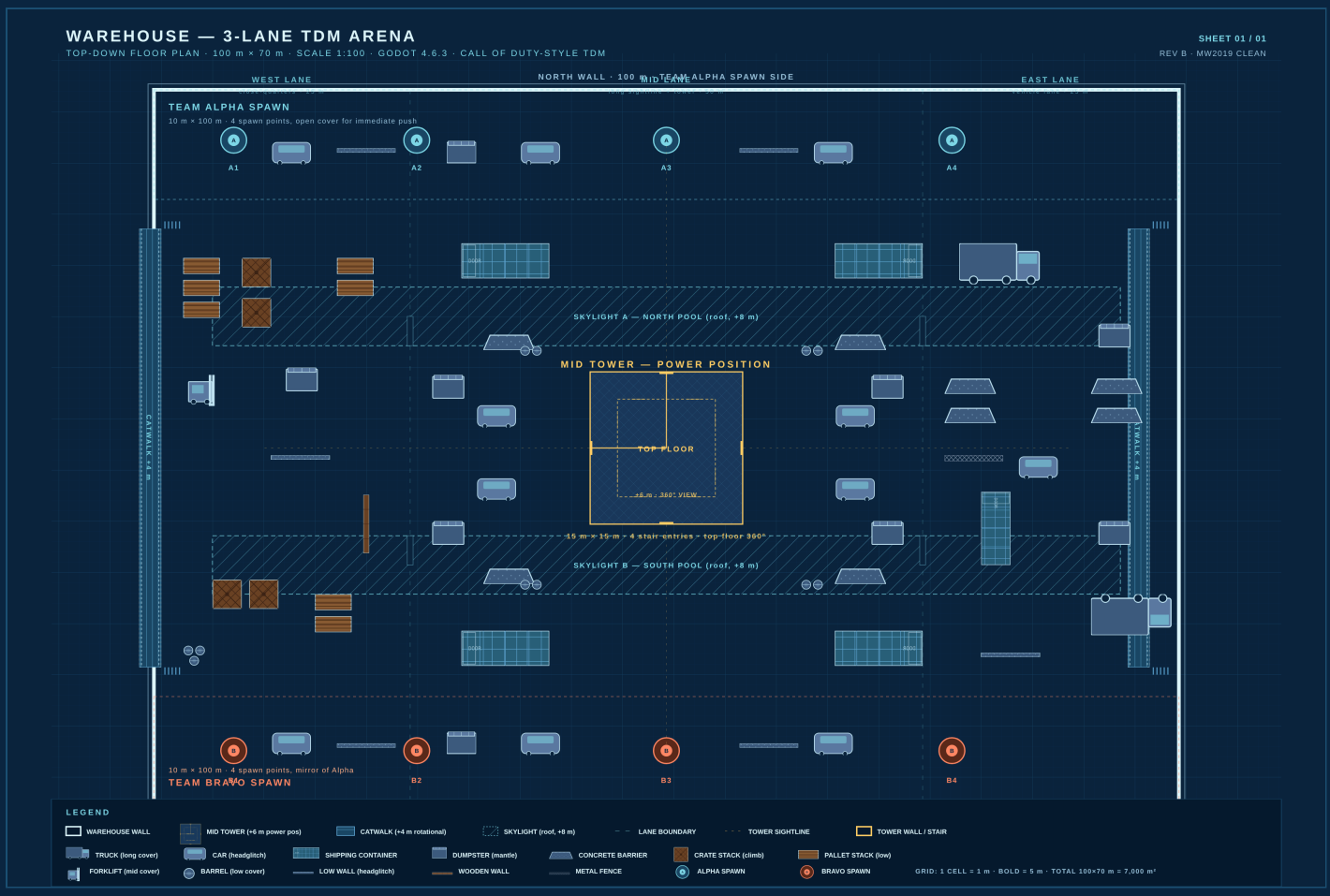
A 100 m × 70 m Call of Duty-style deathmatch map

A huge industrial warehouse rebuilt for run-and-gun TDM flow: classic 3-lane layout (west close-quarters / mid long-sightline / east vehicle lane), symmetric Alpha/Bravo spawns on opposite short sides, a central mid tower as the contested power position, two wall-mounted catwalks for rotational verticality, and a full cover ecosystem tuned for headglitches and mantle play — cars, dumpsters, shipping containers, pallets, crates, forklifts, and concrete barriers.

FOOTPRINT	AREA	INNER KILLHOUSE	LANES
100 × 70 m	**7,000 m²**	**30×15 m sideways**	**3 classic**

TOP-DOWN FLOOR PLAN

Blueprint — 1:100



Cyanotype top-down view · Grid 1 cell = 1 m · Bold line = 5 m · All dimensions in meters

CONCEPT

Map Overview

The map is a single huge warehouse shell, 100 meters wide and 70 meters deep, capped with two long strip skylights that cast bright daylight pools on the action zone. Inside, a classic CoD 3-lane layout runs north-south: the west lane is close-quarters crate-and-pallet aisles, the mid lane is a long open sightline anchored by a central mid tower, and the east lane is a vehicle lane with trucks and concrete barriers. Team Alpha spawns on the north short side, Team Bravo mirrors on the south. Both spawns have four spread points and immediate cover so no one dies in the first two seconds.

The inner killhouse is the map's power position — a 30 m × 15 m building (sideways, long axis east-west) laid out as a maze of 6 rooms in a 2×3 grid, with a big central tower (6 m × 6 m, +6 m) in the dead center. Every room has cover inside (barrels, crates, low walls, pallets, dumpsters) and 2–3 doorways connecting it to adjacent rooms, so no single defender can lock it down. The central tower has 4 entries (N, S, E, W) and gives 360° view from the top. Two catwalks run along the east and west warehouse walls at +4 m, giving rotational verticality — they exist for flanking, not camping.

7,000 m²	3	8	~3 min
TOTAL FLOOR AREA	DISTINCT LANES	SPAWN POINTS (4+4)	AVG TDM MATCH

DESIGN PILLARS

- **3-lane flow.** Each lane has a distinct personality (CQB / long-range / vehicle). Players can commit to a lane or rotate via cross-passages at the north and south action zone.
- **Symmetric balance.** Alpha and Bravo spawns are mirror-imaged. No side gets a shorter rotation to the mid tower.
- **Power position is contestable.** The mid tower has four stair entries and is exposed from the catwalks. Holding it requires team support, not a single camper.
- **Headglitch-friendly cover.** Cars, low walls, and concrete barriers are placed at chest height so players can shoot over them while crouching — the core CoD cover loop.
- **Fast movement routes.** Catwalks and stacked containers provide mantle and climb paths. No dead-ends — every cover cluster has a flank exit.
- **Light as info.** Two skylight strips cast hard-edged pools on the action zone. Players crossing them are visible from the tower top and the catwalks.

GAMEPLAY NOTE

The skylights are open roof frames with safety grating, not glass. They cannot be shot through but they silhouette anyone crossing the light pool below. The central tower (inside the sideways killhouse) is positioned between the two skylight strips so a player holding the tower top is backlit from both spawns. The 30 m building blocks the direct spawn-to-spawn sightline through the mid lane — attackers must go around or through the killhouse maze, creating natural engagement zones on the north and south approaches.

SIX ZONES

Zone-by-Zone Breakdown

1. Team Alpha Spawn — North

10 M × 100 M

Four Alpha spawn points spread across the 100 m width, each behind a car or low wall so players don't die in the first 2 seconds. A dumpster anchors the center for mantle play. The spawn zone is open enough to push into any of the three lanes immediately.

2. West Lane — Close-Quarters Aisles

25 M × 50 M

Crate stacks and pallet rows create winding aisles for SMG and shotgun play. A forklift and dumpster provide mid-height cover, and a wood wall subdivides the lane. Barrel clusters near the south approach give shoot-through cover. Two cross-passages connect to the mid lane — one north, one south — so flankers can rotate without backtracking.

- 3× PALLET STACK
- 4× CRATE STACK
- 1× FORKLIFT
- 1× DUMPSTER
- 1× LOW WALL
- 1× WOOD WALL
- 3× BARREL
- 2× CROSS-PASSAGE

3. Mid Lane — Long Sightline + Tower

50 M × 50 M

The main engagement lane. A 50 m sightline runs from Alpha spawn to the tower's north face, and from the tower's south face to Bravo spawn — the extended 30 m tower blocks direct spawn-to-spawn shots. Two shipping containers flank each approach, with cars and concrete barriers for headglitches. Dumpsters and barrel clusters give mantle points alongside the tower's long walls. Tower holders can shoot both approaches but are exposed from four stair entries and two catwalk bridges.

- 4× SHIPPING CONTAINER
- 6× CAR (HEADGLITCH)
- 4× CONCRETE BARRIER
- 4× DUMPSTER
- 8× BARREL
- 1× MID TOWER (15×30 M)

4. East Lane — Vehicle Lane

25 M × 50 M

Long open sightline for assault rifles and LMGs. Two trucks provide long full-body cover, with concrete barriers creating low headglitch positions between them. A shipping container mid-lane breaks the sightline for flankers. A metal fence partially divides the south approach — vision-piercing only, blocks movement but not bullets. Two cross-passages connect to the mid lane, mirroring the west lane's rotation options.

- 2× TRUCK
- 4× CONCRETE BARRIER
- 1× CAR
- 1× SHIPPING CONTAINER
- 2× DUMPSTER
- 1× LOW WALL
- 1× METAL FENCE
- 2× CROSS-PASSAGE

5. Inner Killhouse — Maze + Central Tower

30 M × 15 M (SIDEWAYS) · 6 ROOMS + CENTRAL TOWER

30×15 m building (long axis east-west) as a killhouse maze: 6 rooms in a 2×3 grid (R1 NW, R2 N-middle, R3 NE, R4 SW, R5 S-middle, R6 SE). A central tower (6×6 m, +6 m) sits dead center with 4 entries (N, S, E, W). 10 internal doorways connect adjacent rooms — every room has 2–3 passages. Every room has cover: R1 barrels, R2 crate, R3 low wall, R4 pallet, R5 dumpster, R6 barrels+crate. Four outer entries (N, S stair + W, E doorway). Central tower = 360° power position, accessible only through the maze.

- 30×15 M BUILDING (SIDEWAYS)
- 6 ROOMS (2×3 GRID)
- CENTRAL TOWER 6×6 M (+6 M)
- 10 INTERNAL PASSAGES
- COVER IN EVERY ROOM
- 4 TOWER ENTRIES (N/S/E/W)
- 360° VIEW FROM TOWER

6. Team Bravo Spawn — South

10 M × 100 M

Mirror of Alpha — four Bravo spawns, same cover. Symmetric; same 2s spawn protection.

MODELING REFERENCE

Cover & Asset List

Every cover piece on the blueprint, numbered by zone. Each row = one Godot scene instance. Cover heights: low (≤1.2 m) = headglitch, mid (1.2–2.0 m) = blocks body, full (≥2.0 m) = blocks all.

ID	ZONE	ASSET	MATERIAL	DIMS (W×D×H M)	NOTES
A-S1..4	Alpha Spawn	Spawn Point ×4	MARKER3D	0.5 × 0.5 × 0.0	Spread E-W, 2s spawn protection
A-C1..3	Alpha Spawn	Car ×3	PAINTED METAL	3.5 × 1.8 × 1.5	Headglitch cover, hood-facing lane
A-L1..2	Alpha Spawn	Low Wall ×2	CONCRETE (PAINTED)	5.0 × 0.4 × 1.2	Chest-height headglitch
A-D1	Alpha Spawn	Dumpster	PAINTED STEEL	2.0 × 1.4 × 1.4	Mantle point, mid cover
W-P1..3	West Lane	Pallet Stack ×3	WOOD	2.4 × 1.0 × 0.8	Low cover, aisle divider
W-CR1..4	West Lane	Crate Stack ×4	PLYWOOD	2.0 × 2.0 × 2.0	Full cover, climbable to catwalk
W-F1	West Lane	Forklift	PAINTED STEEL	2.4 × 1.5 × 2.2	Mid cover with vision gaps
W-D1	West Lane	Dumpster	PAINTED STEEL	2.0 × 1.4 × 1.4	Mantle point
W-L1	West Lane	Low Wall	CONCRETE (PAINTED)	5.0 × 0.4 × 1.2	Headglitch near south approach
W-W1	West Lane	Wood Wall	PLYWOOD ON FRAME	4.0 × 0.2 × 2.4	Sub-divider, full cover
W-B1..3	West Lane	Barrel ×3	PAINTED STEEL	0.6 × 0.6 × 1.0	Low cover, shoot-through
M-CN1..4	Mid Lane	Shipping Container ×4	CORRUGATED STEEL	6.0 × 2.4 × 2.6	Full cover, climbable to catwalk
M-C1..4	Mid Lane	Car ×4	PAINTED METAL	3.5 × 1.8 × 1.5	Headglitch flanking tower
M-B1..4	Mid Lane	Concrete Barrier ×4	CONCRETE (JERSEY)	5.0 × 1.0 × 1.2	Low cover, headglitch
M-D1..4	Mid Lane	Dumpster ×4	PAINTED STEEL	2.0 × 1.4 × 1.4	Mantle points, mid cover
M-BR1..8	Mid Lane	Barrel ×8	PAINTED STEEL	0.6 × 0.6 × 1.0	Low cover clusters
E-T1..2	East Lane	Truck ×2	METAL / GLASS	5.0 × 2.5 × 2.8	Long full-body cover
E-B1..4	East Lane	Concrete Barrier ×4	CONCRETE (JERSEY)	5.0 × 1.0 × 1.2	Low cover between trucks
E-C1	East Lane	Car	PAINTED METAL	3.5 × 1.8 × 1.5	Headglitch mid-lane
E-CN1	East Lane	Shipping Container	CORRUGATED STEEL	6.0 × 2.4 × 2.6	Full cover, breaks sightline
E-D1..2	East Lane	Dumpster ×2	PAINTED STEEL	2.0 × 1.4 × 1.4	Mantle points
E-L1	East Lane	Low Wall	CONCRETE (PAINTED)	5.0 × 0.4 × 1.2	Headglitch near south
E-F1	East Lane	Metal Fence	CHAIN-LINK ON STEEL	5.0 × 0.1 × 2.0	Vision-piercing, bullet-piercing
K-01	Inner Killhouse	Building (30×15 m sideways, 6-room maze)	CONCRETE + STEEL	30.0 × 15.0 × 3.5	2×3 grid, 10 passages
K-02	Inner Killhouse	Central Tower (6×6 m, +6 m)	CONCRETE + STEEL	6.0 × 6.0 × 6.0	Dead center, 4 entries, 360°
K-03..20	Inner Killhouse	Doorways ×10 + Room Cover ×6 + Entries ×4	MIXED	var.	2–3 passages/room; cover in every room
CW-01..06	Catwalks	Wall Catwalk ×2 + Stairs ×4	STEEL GRATING	40.0 × 1.5 / 1.5 × 4.0	+4 m, E/W walls, 2N + 2S access
B-S1..4	Bravo Spawn	Spawn ×4 + Cover	MIXED	var.	Mirror of Alpha

READY-TO-PASTE

AI Image Generation Prompt

Paste the prompt below into Midjourney v6, SDXL, or Flux for an interior concept render. Tuned for the MW2019 clean aesthetic and the 3-lane TDM layout — encodes the warehouse shell, mid tower power position, catwalks, skylights, lane structure, and CoD-style cover ecosystem so the AI produces a usable reference render rather than a generic warehouse.

MASTER PROMPT — WIDE INTERIOR ESTABLISHING SHOT

Midjourney v6 · --ar 16:9 --style raw --v 6

```
Interior wide-angle establishing shot of a massive modern warehouse converted into a Call of Duty-style multiplayer TDM arena, 100 meters by 70 meters, polished concrete floor with faded painted safety lane markings. Two long horizontal skylight strips in the high steel-truss ceiling cast bright pools of natural daylight onto the action zone below. The interior is divided into three parallel lanes running north-south: a west close-quarters lane of stacked wooden pallets and plywood crate aisles, a wide central mid lane anchored by an inner killhouse building in the dead center, and an east vehicle lane with parked delivery trucks and concrete jersey barriers. The inner killhouse is a 30 by 15 meter concrete-and-steel building oriented sideways (long axis east-west), laid out as a maze of 6 rooms in a 2 by 3 grid, with 10 internal doorways connecting adjacent rooms so every room has 2 to 3 passages to its neighbors. A big central tower 6 by 6 meters rises to 6 meters height in the dead center of the killhouse, with stair entries on all four faces giving 360 degree views from the top. Every room has cover inside: barrels, crate stacks, low concrete walls, pallet stacks, and dumpsters. Two narrow steel catwalks run along the east and west warehouse walls at 4 meters height, providing rotational verticality. Scattered cover throughout the lanes: shipping containers, sedans with hoods at chest height for headglitches, dumpsters, forklifts, oil drums, low concrete walls, and chain-link fence panels. Symmetric team spawns on the north and south short walls, each with four spawn points behind car cover. Modern Warfare 2019 clean aesthetic, SWAT training facility, dust motes in the skylight beams, photo-realistic, 35mm lens, eye-level three-quarter angle, cinematic composition, high detail, color graded cool blue-gray with warm skylight highlights, Unreal Engine 5 render quality, 8k.
```

VARIANT PROMPTS

VARIANT A — TOP-DOWN ORTHOGRAPHIC

SDXL · 1536×1024

```
Top-down orthographic floor plan render of a 100 by 70 meter Call of Duty-style multiplayer TDM warehouse map, classic 3-lane layout with west close-quarters crate aisles, central mid lane with a two-story mid tower power position, and east vehicle lane with trucks and barriers. Two long horizontal skylight strips in the roof cast daylight pools on the action zone. Symmetric team spawns on north and south short walls. Cover of shipping containers, sedans, dumpsters, forklifts, oil drums, low concrete walls, and pallets scattered through the lanes. Two steel catwalks run along the east and west walls. Modern Warfare 2019 clean aesthetic, polished concrete floor with painted safety lines, cyanotype blueprint color palette with navy background and cyan-white linework, ultra detailed, isometric lighting from above, video game level design concept.
```

VARIANT B — MID TOWER POV

Midjourney v6 · --ar 16:9 --style raw

```
First-person view from the top of the central tower (6 by 6 meters, 6 meters tall) in the dead center of a 30 by 15 meter inner killhouse building oriented sideways, in the center of a massive Call of Duty-style TDM warehouse arena, looking down 360 degrees through open window frames at the maze of 6 rooms below. The killhouse is a 2 by 3 grid of rooms with 10 internal doorways, every room containing cover. Two skylight strips cast bright daylight pools across the polished concrete floor on the north and south approaches. Cover of shipping containers, sedans, dumpsters, and concrete barriers scattered through the lanes. Steel catwalks run along the east and west walls at 4 meters height. Symmetric team spawns at the north and south ends. Modern Warfare 2019 clean aesthetic, dust in the light beams, photo-realistic, 50mm lens, cool blue-gray color grade with warm skylight highlights, cinematic, Unreal Engine 5 render quality, 8k.
```

IMPLEMENTATION GUIDE

Godot 4.6.3 Scene Notes

Build notes for Godot 4.6.3. The map is one main scene with nested child scenes per zone, so a designer can hide/show zones without unloading the whole warehouse.

Scene Tree Structure

```
Warehouse_Main (Node3D)
├─ Shell/                  Floor (100×70m concrete) + Walls_N/S/E/W (8m steel) + Roof_Trusses
├─ Skylights/              2× Area3D + OmniLight3D + VolumetricFog god-rays
├─ Lanes/
│  ├─ West_Lane/           Cover_Group (pallets, crates, forklift, dumpster, barrels, walls)
│  ├─ Mid_Lane/            Cover_Group (containers, cars, barriers, dumpsters, barrels) + Mid_Tower
│  └─ East_Lane/           Cover_Group (trucks, barriers, container, dumpster, fence)
├─ Inner_Killhouse/        KillhouseBuilding.tscn (30×15m sideways, 6-room maze) + CentralTower.tscn (6×6m, +6m) +
Room_Doorways ×10 + Room_Cover ×6 + Entries ×4
├─ Catwalks/               West_Catwalk + East_Catwalk (+4m) + Stairs ×4 (2N + 2S)
├─ Spawns/
│  ├─ Alpha_Spawn/         SpawnPoint ×4 (team=ALPHA) + Cover_Group
│  └─ Bravo_Spawn/         SpawnPoint ×4 (team=BRAVO) + Cover_Group
├─ Lighting/               WorldEnvironment + DirectionalLight3D (baked) + ReflectionProbe ×4
├─ Navigation/             NavigationRegion3D (baked navmesh, includes tower + catwalks)
└─ TDM_Manager             score, spawn rotation, round timer
```

Geometry — CSG vs Imported Mesh

- **Shell + lane dividers + tower structure:** CSG boxes for greybox — fast iteration, collision auto-generates with `use_collision = true`. Promote to glTF meshes once layout is signed off.
- **Cars, trucks, dumpsters, containers, crates, pallets, forklifts, barrels:** Always imported glTF meshes. Reuse one `Car.tscn` for all 7 cars on the map, one `Container.tscn` for all 5 containers, etc.
- **Inner killhouse:** Two scenes — `KillhouseBuilding.tscn` (30×15 m sideways outer building with 6 rooms in 2×3 grid, 10 internal doorways, 4 outer entries) and `CentralTower.tscn` (6×6 m tower in dead center, +6 m, 4 entries N/S/E/W, 4 top-floor windows, snap Marker3D on top). Each room gets its own cover instance (barrels, crate, low wall, pallet, dumpster).
- **Catwalks:** One `Catwalk.tscn` with the walkway mesh, railing mesh, and stair collider. Instance twice (E and W walls).
- **Skylights:** Open rectangles in the roof mesh + transparent emissive panel below for the light shaft. No glass — open frames with safety grating only.

Collision Layers

- 8 layers: World, Cover, Player, Enemy, Projectiles, Triggers, Navigation, Skylight Shaft. Cover splits into Full (containers/tower/trucks — block all), Low (barriers/cars — headglitch, block body below 1.2m), Shoot-Through (barrels/fences/pallets — block movement only). Bullet raycast respects the split.

TDM Logic, Navigation & Performance

- TDM spawn manager picks spawn by team + min-distance-from-enemy. If all spawns within 20m of enemy, fall back to danger spawn with 2s invulnerability. Symmetric — Alpha and Bravo use the same logic.
- Anti-spawn-trap: 3 lanes fork immediately out of each spawn, so campers cannot lock a single exit. Spawn cover gives headglitch defense without trapping players.
- Mid tower contest: 4 stair entries + 8 window frames. Any player on top is exposed from ≥3 angles. Area3D triggers "Tower Held" callout when one team has 2+ players for 5s.
- Catwalk rotation: 1.5m wide, no cover, waist-high railings. One-way mantle at tower bridge.
- TDM score: +100/kill, +25 for tower hold (per 5s). First to 100 kills or 5-min timer wins.
- Navmesh + Lighting: NavigationRegion3D over 100×70m floor; separate regions on tower top + catwalks via Link3D. VolumetricFog 0.04–0.06; SDFGI low; skylights 3–5× ambient.

FREE CC0 / CC-BY SOURCES

Free Texture Site Guide

All four sites below are free, allow commercial use, and provide PBR texture sets (albedo, normal, roughness, AO, height, metallic). No per-asset crediting required, and most offer bulk downloads so you can pull an entire category at once.

1. Poly Haven

polyhaven.com/textures

CC0 · 8K MAX · PBR FULL SET · FREE FOR COMMERCIAL USE

The gold standard. Every texture is a complete PBR set with up to 8K resolution. Use their **bulk download** page to grab entire categories. For this MW2019 warehouse: concrete (floor + barriers), corrugated metal (walls + shipping containers), plywood (crates + pallets), and steel (catwalks + tower). They also have HDRIs for the skylight lighting setup.

CONCRETE FLOOR | CORRUGATED STEEL WALLS | SHIPPING CONTAINER | PLYWOOD CRATES | STEEL CATWALK | HDRI FOR SKYLIGHTS

2. AmbientCG

ambientcg.com

CC0 · 8K MAX · PBR FULL SET · BULK ZIP DOWNLOADS

Largest CC0 library after Poly Haven. Strong on painted metal (cars, trucks, dumpsters, forklifts, barrels), wood (pallets, crates), and chain-link (fences, skylight grating). Every texture comes as a ZIP with all PBR maps in multiple resolutions (1K/2K/4K/8K) so you can pick the right size per asset.

PAINTED METAL (CARS/TRUCKS) | DUMPSTER STEEL | PALLET WOOD | CHAIN-LINK FENCE | SAFETY GRATING

3. ShareTextures

sharetextures.com

CC0 / CC-BY · 4K-8K · PBR FULL SET · FREE TIER COVERS COMMERCIAL USE

Heavily organized by material category — best site when you know exactly what surface you need. For this map: "painted concrete with yellow safety line" (floor lane markings), "polished concrete with oil stains" (action zone floor), "rusty truck paint" (east lane vehicles), and "industrial steel grating" (catwalk treads). Their industrial and vehicle categories are particularly strong.

PAINTED SAFETY LINES | OIL-STAINED CONCRETE | RUSTY TRUCK PAINT | STEEL GRATING | INDUSTRIAL GRIME

4. Texture Haven (now part of Poly Haven)

polyhaven.com/textures (filter: "industrial")

CC0 · 8K · PBR · SAME ARCHIVE, EASIER CATEGORY FILTER

Texture Haven merged into Poly Haven; the legacy URL redirects. Mentioned separately because the old "Texture Haven" name still surfaces in tutorials. Use the Poly Haven industrial filter to find the corrugated metal, rusty steel, and concrete panels that originally carried the Texture Haven label — these are the highest-quality entries in the archive and are exactly what a Modern Warfare 2019-style warehouse needs. Note: many older tutorials also reference cc0textures.com — that's the former name of AmbientCG (entry #2 above); same library, same license.

RUSTY STEEL PANELS | WAREHOUSE INTERIOR METAL | CONCRETE BLOCK WALLS

RECOMMENDED PACK & GODOT IMPORT

Pull from **Poly Haven** (concrete, corrugated metal, plywood, steel, HDRI), **AmbientCG** (painted metal, wood, chain-link), **ShareTextures** (industrial/safety). Drop into `res://textures/<category>/`, use **StandardMaterial3D** ORM workflow (albedo → *Albedo/Texture*; pack AO/Roughness/Metallic into one ORM; normal → *Normal/Texture*). Enable **VRAM compression** + **mipmaps** on all PBR maps — the 7,000 m² warehouse needs mipmapping for frame rate.